

Milestone B: Detailed Experimental Plan

Instructor Dr. George Anwar

Due Date Friday, March 13th, 2026 at 23:59

Now that you have received feedback on your project proposal from the shop staff and us, we would like you to flesh out the precise sequence of experiments that you plan to do. In doing this we would like you to draw on the framework for design of experiments that was laid out in the lectures.

The below points are meant to give guidance for writing your plan, but not to be unnecessarily prescriptive. If you think you need to use a different structure for your experiments, then feel free to do so, with appropriate justification.

Recognizing that successful experiments tend to involve some iteration, we recommend that you plan for at least two phases to your experimentation, with some analysis (i.e. plotting, interpretation, and discussion) of data between the phases to allow for refinement of the later experiments.

Writing the Experimental Plan

Your proposal must include the following sections. The goal is clarity and feasibility, not length.

- **Purpose and Research Question:** state your measurement goal (what physical quantity you'll estimate or target you want to reach via the measurement process), the primary factors you'll vary, and the key response(s) you'll record. Also state why this is worth measuring. Include testable hypotheses:

Example: We will quantify how laminate thickness and fiber orientation affects UTS and the Weibull Modulus.

H_0 : orientation has no effect on UTS; H_a : orientation affects UTS

- **Timeline and Experimental Method Prototyping:** This phase should occupy roughly 20% of your experimental effort for the project. The idea here is to look in a compact way at all the independent variables you think may be significant, to identify those that have a significant effect, and to eliminate those that do not. It is recommended that you explore a wide range of parameter values in search of an optimal region of the parameter space for later, more detailed, investigation. For this initial phase, it will be particularly valuable to deploy the techniques covered in the lectures including, e.g., 2^{k-1} fractional factorials.

Example: k is the number of factors or variables you are testing. We call something a full factorial test if you are running all 2^k tests but generally speaking, this is expensive and time consuming so we employ a fractional factorial. The 2 means that each factor/variable is tested at 2 levels a minimum and a maximum setting generally.

You would like to see which factors (cutting speed = A, feed rate = B, depth of cut = C, and tool material = D) most affect the tool life in a mill. You don't have time to run all 2^4 tests so you employ a 2^{4-1} design i.e. a half fraction design since 8 is half of 16.

Say we decide to test A, B, C, for us to incorporate the last variable D, we need to then define a relation. Let's define $D = A \times B \times C$, then we can build the design matrix and estimate all 4 parameters. This fractional factorial design will reveal important trends but will have some factor interactions.

- In your plan, explicitly state which variables (factors) you will investigate, and at what values (levels). State whether you will use full or fractional factorial experimental design, and whether you will use a center point to look for curvature in any trends. The screening phase should include some replicate experiments to determine the inherent amount of variability in your method. You may choose to do replicates at every combination of parameter values, for a subset, or just for the center point.
- For this part of your plan, include a table detailing the sequence of runs you will do, with the values of every parameter. Make sure you consider randomization of sequence and blocking of any potential confounding effects in your design.

– Variables and Levels:

- * Responses (with units and how measured): e.g., UTS (MPa) from instron CSV; strain at break from crosshead/gauge; elastic modulus (GPa) from slope/vibration test.
 - * Factors (controllable): pick 2-6 that plausibly move the response (e.g. orientation, thickness/pplies, pressure/crosshead speed/ temperature/specimen length)
 - * Nuisance Factors: environment, operator; plan to block or hold constant.
 - * Constraints: instrument ranges/limits, safety.
- Experimental Design: screening with a fractional factorial. Use a 2^{k-1} fractional factorial to screen main effects efficiently.
- **Interim data analysis phase:** Here, we would like you to think about how you will interpret the data from your screening phase. Please be specific about graphs you will plot, regressions you will compute, and tests you will conduct. If your project has a clear hypothesis, you will definitely want to carry out an appropriate hypothesis test on your screening data, to see what kind of statistical power your screening design has. For example, if you think you see an effect but the null hypothesis of no difference in means is not rejected at the 5% level, that might be a good reason to increase the number of replicates to try to increase the statistical power you can later bring to bear. This is not to say whether or not you will ultimately reject the null hypothesis by obtaining more data, but the idea is to make sure you get enough data to be content that you have tested the hypothesis to the best of your ability. If you are evaluating a range of levels for a particular parameter (e.g. different material formulations), you may also want to use ANOVA to look for trends.
 - If you observe random variability that is comparable to the magnitude of the factor effects you observe, you may need to either (a) try to diagnose sources of this random error before entering the refined phase, or (b) increase the number of replicates in your refined phase so that your sample size goes up and your standard error values are correspondingly reduced.
 - **Refined Experimental Phase:** You are unlikely to know yet precisely which parameters and values you will explore in this phase: you will probably depend on the screening phase to make these decisions. However, you can set out the basic boundaries of how much effort you think is reasonable: e.g. number of tensile tests, whether you will investigate in more detail the strongest effect from your screening experiments, the strongest two effects, or some other number of factors. You could also state, for example, if you plan to implement a central composite design with enough data points to fit a second-order model, or if you will make a decision on the exact experimental design based on whether you see curvature in your screening phase. You could also consider any physical models against which you may want to compare your results.
 - **Comments made by Hesse Shop Staff.** Summarize them and discuss any issues here.

Submission Notes

The experimental plan length is approximate (roughly 3 pages); focus on clarity and feasibility rather than strict page limits. The proposal should demonstrate that your team has thought carefully about what will be measured, how measurements will be made, and how results will lead to meaningful engineering insight. **You must typeset your document in L^AT_EX.** You may use the same template as Milestone A, just change the headings.

You will submit on Gradescope under the **Final Project: Milestone B** assignment. Please remember to add all your group members on the submission.